

SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I NITHISH KUMAR K (192521266) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: knithishkumar

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
The system must handle millions of concurrent users with minimal buffering.
Design a cloud architecture using scalability and caching techniques.
Explain how load balancing and auto-scaling can ensure smooth streaming.
Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers.
They face challenges in maintaining consistency and managing deployments.
Propose a solution using container orchestration and multi-cloud strategies.
Explain how service discovery and scaling can be handled efficiently.
Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
The organization needs a robust backup and disaster recovery strategy.
Design a solution using cloud-native backup, replication, and versioning.
Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.

Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

1. Media Company – Cloud Video Streaming Platform

Cloud Architecture Design

- Host the streaming platform on cloud virtual machines or Kubernetes clusters.
- Store videos in cloud object storage.
- Use a Content Delivery Network (CDN) to cache videos close to users.
- Place a Load Balancer before application servers.
- Enable Auto Scaling to add/remove servers automatically.
- Use a cloud database for user accounts and subscriptions.

Load Balancing and Auto Scaling

- **Load Balancer**
 - Distributes user requests equally among servers.
 - Prevents server overload.
 - Improves response time and availability.
- **Auto Scaling**
 - Automatically increases servers during peak traffic.
 - Removes extra servers during low traffic.
 - Maintains smooth video streaming with minimal buffering.

Cost Optimization Strategies

- Use auto-scaling to avoid paying for idle servers.
- Store old videos in low-cost storage.
- Use CDN caching to reduce bandwidth cost.
- Monitor resource usage and remove unused resources.
- Use reserved instances for predictable workloads.

Conclusion

This architecture provides **high scalability, fast streaming, low latency, and reduced operational cost.**

2. Fintech Startup – Containers and Multi-Cloud

Container Orchestration Solution

- Package applications using Docker containers.
- Deploy containers using Kubernetes.
- Run Kubernetes clusters across multiple cloud providers.
- Use Infrastructure as Code (IaC) for consistent deployment.

Service Discovery and Scaling

- Kubernetes Service automatically discovers application services.
- Internal DNS allows containers to communicate easily.
- Horizontal Pod Autoscaler increases or decreases pods based on CPU or memory usage.

Benefits of Multi-Cloud

- High availability.
- Avoids vendor lock-in.
- Better disaster recovery.
- Better performance by using the nearest cloud region.

Risks

- More complex management.
- Higher monitoring effort.
- Increased security configuration.
- Possible higher networking costs.

Conclusion

Container orchestration with Kubernetes ensures **consistent deployment, scalability, and high availability** across multiple cloud providers.

3. E-Learning Platform – Backup and Disaster Recovery

Backup and Recovery Design

- Enable automatic cloud backups.
- Use object storage versioning.
- Replicate data to another cloud region.
- Schedule regular snapshots of databases and virtual machines.
- Test backup restoration periodically.

Achieving RTO and RPO

Recovery Time Objective (RTO)

- Maintain standby servers.
- Use automated failover.
- Quickly restore data from backups.

Recovery Point Objective (RPO)

- Perform frequent backups.
- Use continuous data replication.
- Enable database synchronization.

Business Continuity Steps

1. Enable automatic backups.
2. Store backup copies in another region.
3. Monitor system health continuously.
4. Perform disaster recovery testing.
5. Automatically switch users to backup servers during failures.

Conclusion

Cloud-native backup and disaster recovery ensure **minimal data loss, fast recovery, and uninterrupted learning services.**

4. Government Agency – Hybrid Cloud

Hybrid Cloud Architecture

- Keep sensitive data in the on-premises data center.
- Store non-sensitive applications in the public cloud.
- Connect both environments using VPN or dedicated private connection.
- Secure communication using encrypted channels.
- Use centralized management tools.

Identity Management

- Implement Single Sign-On (SSO).
- Use Multi-Factor Authentication (MFA).
- Apply Role-Based Access Control (RBAC).
- Maintain centralized user authentication.

Encryption

- Encrypt data during transmission using TLS.
- Encrypt stored data using AES-256.
- Secure encryption keys using a Key Management Service (KMS).

Challenges

- Complex infrastructure management.
- Data synchronization issues.
- Compliance requirements.
- Higher maintenance cost.
- Security policy management across environments.

Conclusion

A hybrid cloud provides **security, compliance, flexibility, and efficient resource utilization.**

5. SaaS Provider – High Availability and Fault Tolerance

Multi-Region Deployment Strategy

- Deploy application servers in multiple cloud regions.
- Use global load balancing.
- Maintain active-active or active-passive architecture.
- Store application data in replicated cloud databases.

Data Replication and Failover

- Continuously replicate data between regions.
- Configure automatic failover.
- Redirect users to the healthy region if one region fails.
- Perform regular failover testing.

Monitoring and Alerting

- Monitor CPU, memory, storage, and network usage.
- Set alerts for high resource utilization.
- Monitor application logs continuously.
- Use cloud monitoring dashboards.
- Send alerts through email or SMS when failures occur.

Conclusion

Multi-region deployment with automatic failover provides **high availability, fault tolerance, and uninterrupted service even during regional failures.**

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented